

This SRFF is symbolically shown as

Flip Flops

We have seen diff. FF's.

- 1) SRFF
- 2) JKFF
- 3) DFF
- 4) TFF
- 5) MS-FF

SRFF is set-reset FF. whose TF is as below.

TT/Charac. table :- Which gives operational charac. of FF.

S	R	Q_{n+1}
0	0	Q_n
0	1	0
1	0	1
1	1	?

Now to obtain charac. eq. of this FF, we have to modify its TT as below :-

S	R	Q_n	Q_{n+1}	
0	0	0	0	} Q_n
0	0	1	1	
0	1	0	0	} 0
0	1	1	0	
1	0	0	1	} 1
1	0	1	1	
1	1	0	?	} ?
1	1	1	?	

$$2^{n-1} \leq N < 2^n$$

$$2 < 8$$

where $Q_n \rightarrow$ o/p of FF before clk pulse.
 $Q_{n+1} \rightarrow$ " " after a clk pulse.

Next step is form a K-map (3 var. K-map) to get an exp. for next state Q_{n+1} .

SR	00	01	11	10
Φ_n	0 1	2 3	X X	4 5

$$\Phi_{n+1} = S + \bar{R} \cdot \Phi_n$$

is charac. eq. of SRFF.

excitation table →

similarly we can find charac. eq. of all other's.

JFF:

K	Φ_n	Φ_{n+1}	
0	0	0	} Φ_n
0	1	1	
1	0	0	} 0
1	1	0	
0	0	1	} 1
1	0	1	
1	1	0	} $\bar{\Phi}_n$
1	1	1	

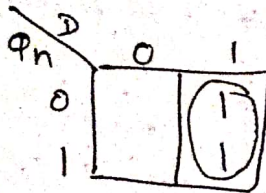
JK	00	01	11	10
0			1	1
1	1			1

$$\Phi_{n+1} = J\bar{\Phi}_n + \bar{K} \cdot \Phi_n$$

J	K
0	0/1
1	0/1
0/1	1
0/1	0
0	X
1	X
X	1
X	0

D-FF

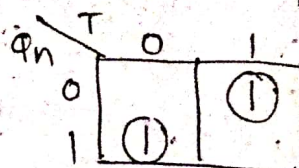
D	Q_n	Q_{n+1}
0	0	0
0	1	0
1	0	1
1	1	1



$$Q_{n+1} = D$$

T-FF

T	Q_n	Q_{n+1}
0	0	0
0	1	1
1	0	1
1	1	0



$$Q_{n+1} = T\bar{Q}_n + \bar{T}Q_n$$

Excitation Table $\rightarrow$ Just now we have seen that by knowing the present state (Q_n) & i/p cond. we can predict the next state (Q_{n+1}). But in design of seq. ckt we usually come across situation in which present state & next state of ckt. (Q_n & Q_{n+1}) are known and we have to find the i/p conditions, that causes this transition of states.

Consider a S-FF whose present & next state are given:

Q_n	Q_{n+1}	S	R
0	0	0	0/1
0	1	1	0
1	0	0	1
1	1	0/1	0

Here $Q_n=0$, & $Q_{n+1}=0$, i.e. o/p does not change when clk pulse is applied. This is true when $S=R=0$, or $S=0$ & $R=1$.

Thus for above cond.ⁿ S should be zero, & R can be 1/0 i.e. X.

Similarly $Q_n = 0 \rightarrow Q_{n+1} = 1$ only when $S = 1$ & $R = 0$.
 $Q_n = 1 \rightarrow Q_{n+1} = 0$ when $S = 0, R = 1$.
 Finally $Q_n = 1 \rightarrow Q_{n+1} = 1$ when $S = R = 0$ or $S = 1, R = 0$.
 i.e. $S = X$ & $R = 0$.

So our table becomes

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

This table/tabulatⁿ of these cond.ⁿ is known as excitation table.

JKFF :-

Q_n	Q_{n+1}	J	K	ie	J	K
0	0	0	0/1		0	X
0	1	1	0/1		1	X
1	0	0/1	1		X	1
1	1	0/1	0		X	0

When $Q_n = 0 \rightarrow Q_{n+1} = 1$ i.e. either $J = 1, K = 0$ or $J = K = 1$ toggles.

D-FF :-

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

TFF

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

⑤

Excitation table of all FF's.

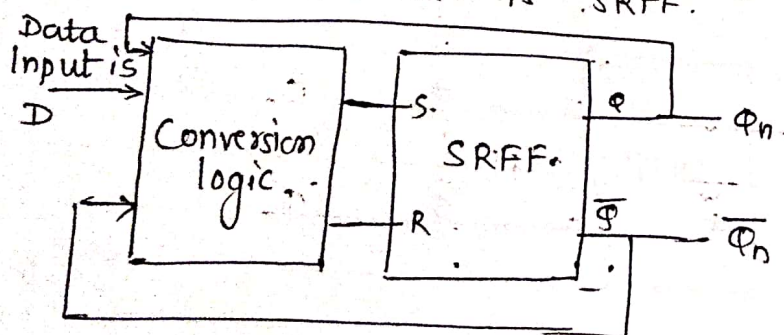
Q_n	Q_{n+1}	S	R	J	K	D	T
0	0	0	X	0	X	0	0
0	1	1	0	0	X	1	1
1	0	0	1	X	0	0	0
1	1	X	0	X	0	1	1

This table is very useful & v. imp design aid for sequential ckts.

Conversion of FF :- In the design of clocked sequential ckts. we come across conversion of 1 FF into the other. For this we req. excitation table and some additional gates/extra connection.

① Suppose SRFF into DFF.

* Basic FF available is SRFF.



* Converted to D means data i/p is D.

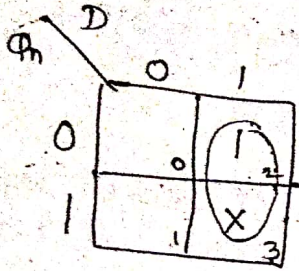
* To get D from SRFF in betⁿ we req. an extra conversion logic, which makes SRFF behaves according to D-FF.

* Now take Excitation table for both FF. Make TT with data i/p D, ~~pre~~ & Q_n as i/p. And find exp. of conversion logic/ for the i/p's of basic FF SRFF.

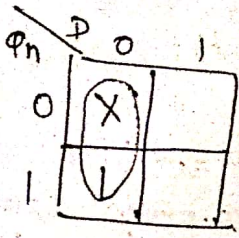
Input Pre state Nxt state FF i/p or o/p of conversion logic.

D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	0
1	0	1	1	0
1	1	1	X	0

D	Q_n	Q_{n+1}	S	R
0	0	0	0	X
0	1	0	0	0
1	0	1	1	0
1	1	1	X	0

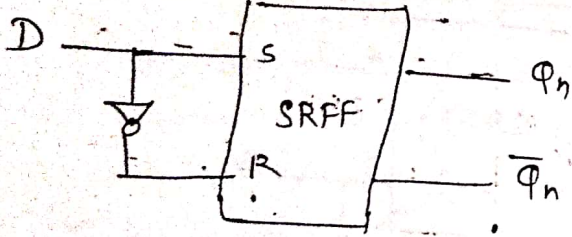


$$S = D$$



$$R = \overline{D}$$

$\therefore$ Ckt becomes



② SRFF to JKFF

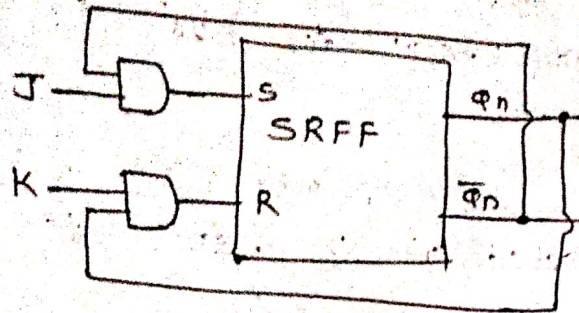
J	K	Q_n	Q_{n+1}	S	R
0	0	0	0	0	X
0	0	1	1	X	0
0	1	0	0	0	X
0	1	1	0	0	1
1	0	0	1	1	0
1	0	1	1	X	0
1	1	0	1	1	0
1	1	1	0	0	1

$\phi_n \backslash JK$	00	01	11	10
0			1	1
1	X			X

$\phi_n \backslash JK$	00	01	11	10
0	X	X		
1		1	1	

$$S = J\bar{\phi}_n$$

$$R = K\phi_n$$

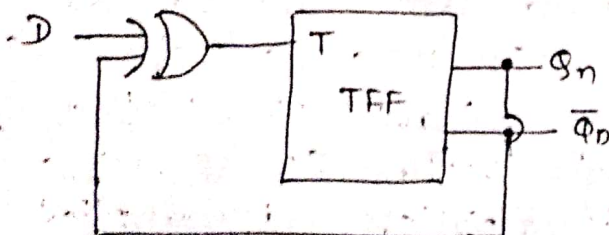


③ T FF to DFF :-

D	ϕ_n	ϕ_{n+1}	T
0	0	0	0
0	1	0	1
1	0	1	1
1	1	1	0

$\phi_n \backslash D$	0	1
0		1
1	1	

$$T = D\bar{\phi}_n + \bar{D}\phi_n = D \oplus \phi_n$$

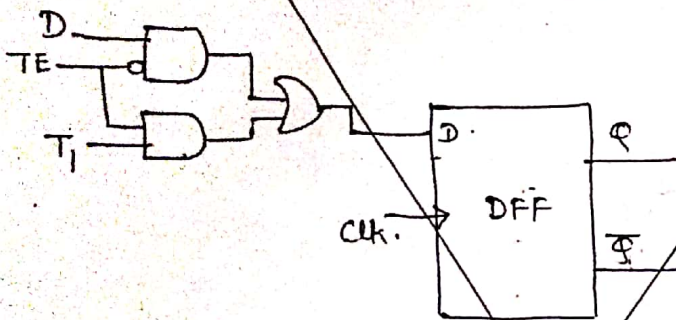


FF Conversion steps :-

- * Take new FF ilp's as ilp and Q_n as ilp.
- * For new FF ilp & Q_n conelⁿ write Q_{n+1} .
- * From Q_n & Q_{n+1} now see the excitedⁿ table of basic FF.
- * Solve using K-map for basic FF inputs.

-----x-----

Scan FF :- It is a DFF with 2 ilp multiplexer connected at D input. As shown



TE — test enable

T1 → test input.

A 2:1 mux is connected to D ilp of FF. So when $TE = 0$, (negated), ckt. behaves as normal DFF. But with $TE = 1$, it goes into test mode. When $TE = 1$, TE is asserted, it takes data from T1, test ilp terminal instead of D-input.

The extra ilp's are used to connected all of an ASIC's FF in a scan chain for testing purpose. Now for testing along with additional logic, $TE = 1$, then test vector bits (pattern) is applied to T1 input, and now with res. to.

each clk pulse, TI ilp's are scanned/shifted into the FF's. Then make TE=0, ie TE is negated and ckt is allow to run for 1/more clk ticks/pulse.

Then the new state of ckt., represented by new values in the FF's is scanned out at Q by asserting TE.

TT for scan FF is

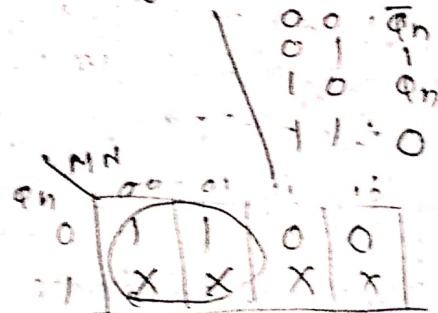
TE	TI	D	clk.	Q	$\bar{Q}$
0	x	0	1	0	1
0	x	1	1	1	0
1	0	x	1	0	1
1	1	x	1	1	0
x	x	x	0	last Q	last $\bar{Q}$
x	x	x	1	last Q	last $\bar{Q}$

Q_n	Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

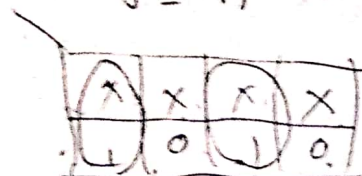
Del-c4 (8 marks)

Q. Design MNFF using JKFF for following TJ: MN, Q_n , Q_{n+1}

M	N	Q_n	Q_{n+1}	J	K
0	0	0	1	1	x
0	0	1	0	x	1
0	1	0	1	1	x
0	1	1	1	x	0
1	0	0	0	0	x
1	0	1	1	x	0
1	1	0	0	0	x
1	1	1	0	x	1

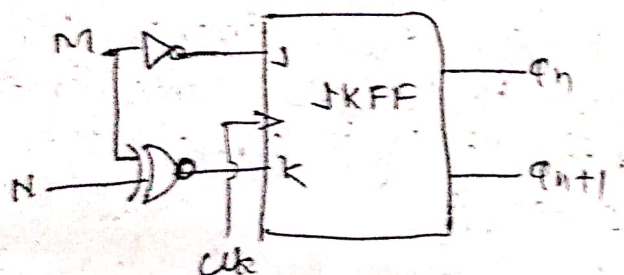


$$J = \bar{N}$$



$$K = \bar{M}\bar{N} + MN$$

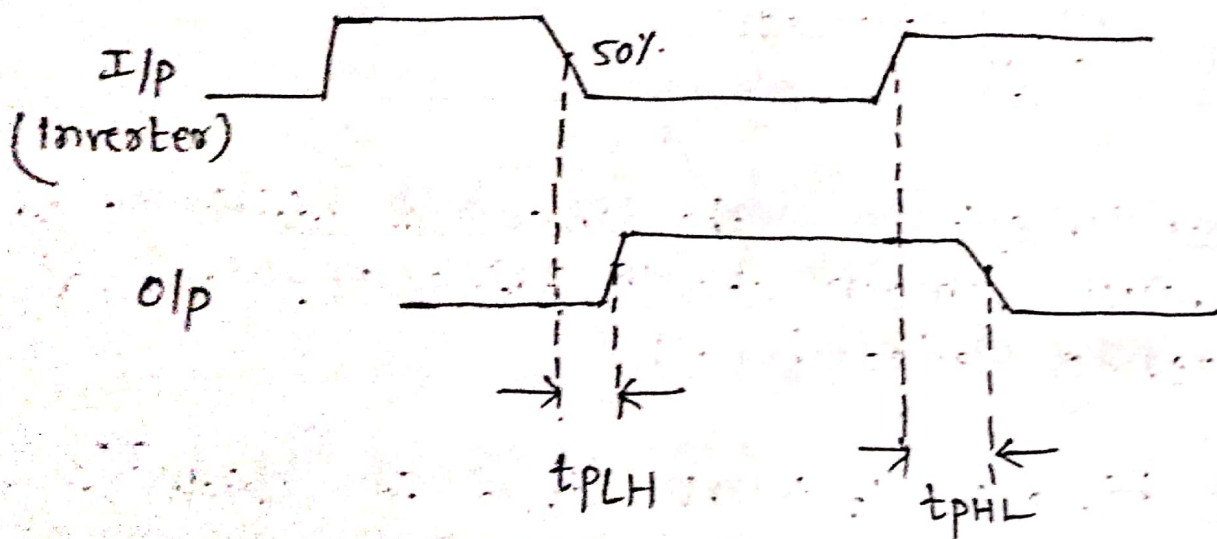
$$= \overline{M \odot N}$$



Propagation delay $\rightarrow$ Is the time req. to change o/p after application of i/p. It is time betⁿ i/p sig. and o/p signal. Ideally should be zero. But we come across two prop. delay times.

t_{PLH} $\rightarrow$ is delay when o/p changes from L to H.

t_{PHL} $\rightarrow$ when o/p changes from H to L.



FF Metastability $\rightarrow$ Any FF have two states, logic 0 & 1 both are stable states. Under normal cond.ⁿ when set up & hold timing para.

Application of FF →

① Switch debouncer / Bounce eliminatⁿ switch.

When mechanical switches are used in digital sys. as i/p device, by which digital info./data 1/0 can be entered into sys. Then there is very serious prob. of switch bouncing.

As switch is pressed/throw from 1 posi. to another, it bounces several times and then come to rest posi. of contact.

Now in sequential ckts., if 1 is to be entered thr' switch, then switch is thrown to corresponding posi. Now as soon as it is thrown it oscillates betⁿ 0 & 1 for some time due to bouncing (make & break) of switch at contact pt. & then come to rest i.e. 1.